

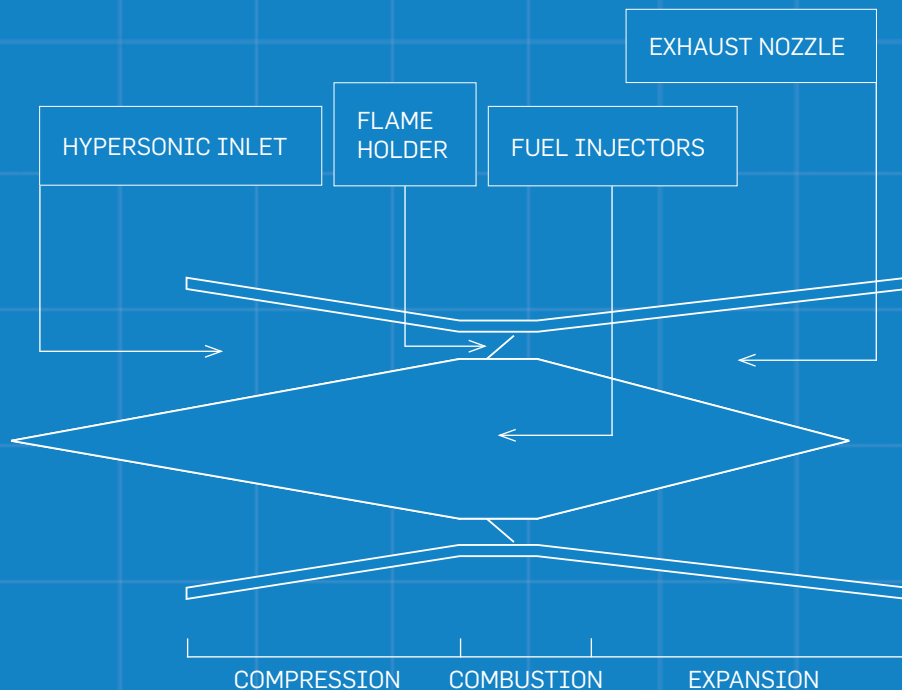
JET PROPULSION

An examination of the various types of jet engines currently in use throughout the world

TYPE	SCRAMJET
SPEED	Mach 3 - 25

The simplest form of a jet engine, a supersonic combustion ramjet (SCRAMJET) is also the fastest air-breathing engine in existence. They are hypersonic engines, i.e., they can travel at speeds of over Mach* 5. These engines are incapable of working at low speeds and are only efficient at a speed of around Mach 6. At low speeds (below Mach 2), the air speed is just not enough for efficient compression and hence, doesn't get hot enough to ignite the fuel.

The way it functions is very simple, air enters through the inlet and gets compressed due to the nozzle design and its speed. This compression heats the air up and at the point of maximum compression, the fuel is injected into the stream. This fuel ignites, rapidly expands, and then hurtles out the exhaust at high speed, thus providing thrust.



TYPE	RAMJET
SPEED	MACH 0.5 - 4

RAMJETs are low-speed variants of the SCRAMJET engine. By low-speed we mean "only" supersonic. The working of these engines is identical to the above mentioned SCRAMJET- the only difference being the inlet, which is designed to compress the air at comparatively high subsonic velocities.

These engines work most efficiently at speeds of around Mach 3 and are not capable of providing thrust below Mach 0.5 without extensive modifications. The SR-71 blackbird uses a combined TURBO-RAMJET engine to allow for subsonic flight as well as for a supersonic cruising speed of Mach 3+.

